

Device Bravo — User Manual

1. Overview

Device Bravo is a ruggedised outdoor sensor hub. It is rated IP67 and is designed for harsh environments including manufacturing floors, outdoor enclosures, and refrigerated facilities.

2. Specifications

Parameter	Value
Operating temperature	-40 °C to 85 °C
Storage temperature	-50 °C to 95 °C
Ingress rating	IP67
Power input	24 V DC PoE + +
Connectivity	Wi-Fi 6, Ethernet, LTE Cat-M1

The **maximum supported temperature for Device Bravo is 85 °C**. Sustained operation above this threshold will trigger thermal throttling at 80 °C and an emergency shutdown at 90 °C.

3. Recommended Firmware

The current **recommended firmware version for Device Bravo is v3.2.1**, released 2025-09-04. This release includes:

- A fix for the LTE roaming reconnect loop reported in v3.1.x.
- Improved deep-sleep accuracy (drift reduced from ± 60 s/day to ± 5 s/day).
- TLS 1.3 support for outbound MQTT.

Devices on firmware v2.x must upgrade to v3.0.0 first, then to v3.2.1. Direct upgrades from v2.x to v3.2.1 are not supported.

4. Calibration

Device Bravo's onboard temperature and humidity sensors should be re-calibrated annually. Use the **Calibration** → **Quick Cal** wizard in the web console with a known reference probe.

5. Mounting

Mount Device Bravo with the antenna pointing up; minimum 30 cm clearance from any metal surface for full Wi-Fi range.